MIL-T-21330E(GL)
19 August 1977
SUPERSEDING
MIL-T-21330D(GL)
18 April 1969

MILITARY SPECIFICATION

TREATMENT: INSECT RESISTANT, FOR PAPER

This specification is approved for use by Natick R&D Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the treatment of paper for protection against insects (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-I-508	-	Insecticides, DDT-Allethrin Insecticides, DDT-Pyrethrin
O-P-921	-	Pyrethrum Extract
VV-K-220	-	Kerosene, Deodorized

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6840

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directd by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Journal of the Association of Official Agricultural Chemists Volume 47

April 1964

(Application for copies should be addressed to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, DC 20044).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

3.1 Materials.

3.1.1 Paper. The specification governing the paper to be treated under this specification shall be cited in the applicable procurement document (see 6.2).

3.1.2 Pyrethrum extract. The pyrethrum extract shall conform to the requirements of O-P-921.

3.1.3 Piperonyl butoxide. The piperonyl butoxide shall conform to the requirements of O-I-508.

3.1.4 Allethrin. The allethrin shall conform to the requirements of O-I-508.

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3.1.5 Sulfoxide. The sulfoxide shall contain a minimum of 88 percent by weight of 1, 2-methylenedioxy-4- [2-octylsulfinyl propyl] benzene when tested as specified in 4.2.1.1.

3.2 Treatment formulation. The treatment formulation shall consist of one of the combinations listed in table I, plus suitable bonding agents such as talc, clay, starch and wetting agents thoroughly dispersed in a liquid phase. If petroleum fractions, oils or resins are used in the formulation, they shall be of the nonoxidizing type when tested as specified in 4.2.1.1.

3.3 Treatment of paper. The paper shall be uniformly treated to conform to the actual concentration requirements specified in table I when tested as specified in 4.3.2.

TABLE I. Concentration of treatment on paper of pure ingredients

	Milligrams per square foot ($0.09290\ m^2$) Minimum
Combination of	
pyrethrins	4
piperonyl butoxide	40
Combination of	
pyrethrins	5
sulfoxide	44
Combination of	
allethrin	10
piperonyl butoxide	80
Combination of	
allethrin	10
sulfoxide	88

3.4 Identification marking of treated paper. All rolls of insect resistant treated paper, except that to be converted into multiwall paper shipping sacks (see 3.4.1) shall be marked INSECT RESISTANT TREATED and the date of treatment (month and year). The lettering shall be at least 1/2 inch (13 mm) high and shall be centered in width of roll and repeated each 2 to 3 feet (0.6 to 0.9 m) of the entire length of the roll. The roll shall be further identified by means of a tag or sheet of paper secured to the core of the roll. The following information shall be legibly printed on the tag or sheet of paper:

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Caution. Chemically treated paper. Separate from any subsistence item by at least two plies of untreated paper (or any material which can be shown to be a barrier to the treatment chemicals). The treated side of the paper should be on the exterior surface of the wrap or pack.

3.4.1 Identification marking for use as a component of multiwall paper shipping sacks. When insect resistant treated paper is to be used as a component of multiwall paper shipping sacks, the treated rolls shall be identified by means of the following "Caution" notice. This caution notice shall be legibly printed on a tag and the tag shall be securely attached to the core of the roll. Also, the caution notice shall appear at the beginning of the roll.

Caution. Insect resistant treated paper with date of treatment (month and year). Separate from any subsistence item by at least two plies of untreated paper (or any material which can be shown to be a barrier to the treatment chemicals). The treated side of the paper should be on the exterior surface of the wrap or pack. Keep tightly rolled and store in cool dark place.

* 3.4.2 Identification of treatment formulation and water repellent. Each shipment of treated paper shall indicate the specific insecticide that has been used. In addition, if a water repellant has been added, it also shall be identified.

3.5 Workmanship. The finished insect resistant treated paper shall have no odor of petroleum or other solvent. The treated paper shall not exhibit a greater deterioration in physical properties than that permitted by the end item specification (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase document.

* 4.2.1.1 Testing of components. Each lot of sulfoxide shall be tested in accordance with the Journal of the Association of Official Agricultural Chemists, Volume 47, April 1964. The lot size shall be expressed in units of gallons (3.8 liter) of sulfoxide. The sample unit for testing shall be a one pint (454 ml) composite obtained by thoroughly mixing equal portions from samples selected at random throughout the lot in the frequency indicated below. The composite shall be placed in a clean, dry, sealed pint (454 ml) container. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, weighing and storage. Duplicate determinations shall be made on the composite and the average value reported to the nearest one percent. The test report shall contain the individual values utilized in expressing the final result. The lot of sulfoxide shall be unacceptable if the composite fails to meet the test requirement specified. When petroleum fractions, oils or resins are used in the formulation, the contractor shall submit a certificate of compliance to the contracting officer certifying that they are of the nonoxidizing type (see 3.2).

Lot size (gallons) (3.8 liter)	Frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The insect resistant treated paper shall be examined for the defects listed herein. The inspection level shall be level S-2 and the acceptable quality level (AQL) shall be 6.5, expressed in defects per hundred units. The lot size shall be expressed in rolls of treated paper. The sample unit shall be one roll of treated paper.

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Examine	Defect
Marking (of paper not for use as a component of multiwall paper sacks)	Omitted, illegible, incorrect, incomplete or not in accordance with requirements. Legend INSECT RESISTANT TREATED with date and year, missing; less than 1/2 inch (13 mm) high. Not approximately in center of width. Not repeated each 2 or 3 feet (610 or 914 mm) of the entire length of the roll. Tag or sheet of paper containing caution notice not as specified or missing.
Marking (of paper as a component of multiwall paper shipping sacks)	Caution notice omitted; illegible, incorrect, incomplete or not in accordance with requirements. Caution notice not securely attached to core of roll. Caution notice does not appear at the beginning of the roll.

* 4.3.2 Testing of treated paper. Each lot of treated paper shall be tested as indicated in table II for the applicable test characteristics specified therein. The lot size shall be expressed as rolls taken from a lot or batch of paper and treated with a single batch of one of the formulations specified in table I. The sample unit for testing shall be three square feet (0.28 m^2) of treated paper and three square feet (0.28 m^2) of paper prior to treatment. The sample units shall be randomly selected. The sample size shall be as shown below. Care shall be exercised to prevent contamination or alteration of the sample units during sampling and testing. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if any sample unit fails to meet the test requirements specified. Prior to testing for concentration of insecticide, the water repellent treatment (if used) shall be removed.

Lot size (rolls)	Sample size (number of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Combination of pyrethrins	Table I	4.4.1			Average of 2		0.1 mg		
piperonyl butoxide	Table I	4.4.2			Average of 2		0.1 mg		
Combination of pyrethrins	Table I	4.4.1			Average of 2		0.1 mg		
sulfoxide	Table I	4.4.3			Average of 2		0.1 mg		
Combination of allethrin	Table I	4.4.1			Average of 2		0.1 mg		
piperonyl butoxide	Table I	4.4.2			Average of 2		0.1 mg		
Combination of allethrin	Table I	4.4.1			Average of 2		0.1 mg		
sulfoxide	Table I	4.4.3			Average of 2		0.1 mg		

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4.4 Test procedures.

4.4.1 Method of analysis of pyrethrins or allethrin on treated paper.

4.4.1.1 Reagents.

Sodium sulfide reagent - Dissolve 1.43 g of A.C.S. grade sodium sulfide ($Na_2S \cdot 9H_2O$) in distilled water and dilute to 100 milliliters. Prepare fresh weekly.

Acetone - A.C.S. grade.

Sodium chloride solution - Dissolve 200 g of A.C.S. grade sodium chloride in distilled water and dilute to four liters.

Petroleum ether - A.C.S. grade, boiling range 30° to 60°C.

Ethyl alcohol - U.S.P., 95 percent.

Hyflo Super-Cel, or equivalent diatomaceous filtering aid.

Cotton - Absorbent grade, U.S.P.

Pyrethrum Extract - 20 percent pyrethrins conforming to O-P-921.

Standard solution of pyrethrins - Weigh 0.1000 g of pyrethrins using the pyrethrum extract and dilute to 1000 ml with ethyl alcohol (1 ml = 0.1 mg of pyrethrins).

Standard solution of allethrin - Weigh 0.1000 g of alpha-dl-trans allethrin (crystalline) and dilute to 1 liter with ethyl alcohol (1 ml = 0.1 mg of allethrin).

4.4.1.2 Apparatus.

Soxhlet extraction apparatus with all glass connections (condensers, Soxhlet extractor, and 250+ml Erlenmeyer flask).

Separatory funnels, 250 ml capacity.

Test tubes, 25 by 200 mm.

Cylindrical funnels approximately 35 by 100 mm.

Water bath, maintained at 70° ± 5°C.

Water bath, maintained at 25° ± 5°C.

Stopwatch or clock with sweep-second hand.

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Photoelectric colorimeter - Klett-Summerson, model 900-3 with green filter having transmittance peak at 540 mu. Use calibrated test tubes of 12.5 mm diameter. (Any comparable colorimeter may be used.)

4.4.1.3 Preparation of standard curve. Pipette aliquots of the standard solution of pyrethrins or allethrin into individual separatory funnels each containing 150 ml of sodium chloride solution. Use 5, 10, 20, 30 and 50 ml aliquots equivalent to 0.5, 1.0, 2.0, 3.0 and 5.0 mg respectively of the pyrethrins or allethrin. Add 15 ml of petroleum ether. Shake for 1 minute, releasing the pressure at intervals. Allow the mixture to separate. Draw off the aqueous layer into another separatory funnel and extract in the same manner using 5 ml of petroleum ether. Repeat the extraction with another 5 ml portion of petroleum ether in a third separatory funnel and then discard the aqueous layer. Combine the extracts (using a minimum of petroleum ether for transfer) into the first separatory funnel. Wash the combined extract with 15 ml of distilled water and draw off the lower layer. Allow the extract to drain into a large test tube through a cylindrical funnel containing a plug of cotton previously wet with petroleum ether. Rinse the separatory funnel with about 10 ml of petroleum ether and pour through the cotton again until all the extract is transferred to the test tube. If necessary, rinse the cotton again until all the extract is transferred to the test tube. The volume of the final extract should be 50 to 60 ml. Add a glass bead to the tube and carefully evaporate the solvent from the extract by slowly immersing the tube in the 70°C water bath, while agitating the tube to initiate bouncing of the glass bead to insure smooth boiling of the solvent. Evaporate the solution down to 1 to 2 ml. Remove the tube from the water bath and evaporate to dryness by carefully blowing a current of nitrogen gas into the tube held in the palm of the hand. If a series of runs are made, warming in a 40°C water bath is more convenient. Pipette 10.0 ml of ethyl alcohol into the tube and swirl to completely dissolve the residue. To the test tube containing the alcohol solution (10.0 ml), add by pipette 1.0 ml of sodium sulfide reagent, start the stopwatch, and mix by gently slapping the tube with the fingers. Do not use a stopper. Place the tube at once in a water bath (70° $\pm$ 5°C). At exactly 15 minutes, by the stopwatch, add 0.3 g of Hyflo Super-Cel, or equivalent, and agitate by vigorous slapping of the tube and immerse at once in the 25° $\pm$ 5°C water bath. At 20 minutes, by the stopwatch, remove the tube and transfer contents directly through rapidflow fluted filter paper into a calibrated colorimeter tube. At exactly 25 minutes, take the transmittance reading. Take all readings at 540 mu exactly 25 minutes after the sodium sulfide reagent is added. Since the Klett-Summerson colorimeter is equipped with a logarithmic scale, the standard curve using concentration in milligrams of pyrethrins is plotted against the dial reading. For other instruments not so equipped, concentration is plotted against log 1/T. Straight-line curves should be obtained. Prepare a reagent blank by pipetting 1.0 ml of sodium sulfide reagent into 10.0 ml of ethyl alcohol and carrying out the color procedure as described above. No visible color should develop. Make all readings with reference to this reagent blank.

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4.4.1.4 Determination of pyrethrins or allethrin on treated paper. Cut into suitable size strips, 0.5 square foot (0.047 m^2) of the paper sample which is free of any colored markings, and place directly in the Soxhlet extractor and extract for 3 hours with 125 ml of acetone. Without transferring from the extraction flask, concentrate the extract on the steam bath to a volume of about 50 ml. Transfer the extract (rinsing with acetone) to a separatory funnel containing 150 ml of sodium chloride solution. From this point, proceed as in the preparation of the standard curve. Paper of the same origin as the sample, but not treated with insecticide, should be run through the procedure. The transmittance readings obtained with the untreated paper should be calculated as milligrams of pyrethrins or allethrin per 0.5 square foot (0.047 m^2) (from the standard curve) and used as the correction blank.

4.4.2 Method for analysis of piperonyl butoxide on treated paper.

4.4.2.1 Reagents.

Acetone - A.C.S. grade.

Kerosene, deodorized - Conforming to VV-K-220.

Standard solution - Prepared from a mixture of pure piperonyl butoxide and 20 percent pyrethrum extract at a ratio of ten parts of pure piperonyl butoxide to one part of pyrethrins by weight to contain 40 mg of pure piperonyl butoxide per 100 ml.

Ethyl acetate - A.C.S. grade, 85 percent

Glacial acetic acid - A.C.S. grade.

Phosphoric acid - A.C.S. grade, 85 percent.

Tannic acid reagent - Dissolve 0.025 g of purified tannic acid in 20 ml of glacial acetic acid by shaking at room temperature until the tannic acid is all in solution and mix this solution with 80 ml of phosphoric acid. This reagent shall be prepared from tannic acid that will give a clear blue color with piperonyl butoxide and remain free of turbidity. The tannic acid shall be purified as follows: To 20 g of reagent grade tannic acid giving a blue color reaction with piperonyl butoxide, add 100 ml of ethyl acetate and agitate the mixture with a mechanical stirrer for 1 hour. Filter by suction through a sintered glass funnel and wash the residue with three 5 ml portions of ethyl acetate. To the combined

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filtrate and washings, add 2 g finely powdered decolorizing carbon and agitate with a mechanical stirrer for one-half hour. Filter by gravity through a double thickness Whatman No. 1 filter paper, or equivalent, into a graduated dropping funnel. Wash residue several times with ethyl acetate from a wash bottle until the volume of the filtrate and washings is about 125 ml. Place the dripping funnel over a 1-liter, 3-neck round bottom flask equipped with a mechanical stirrer and add the filtrate dropwise to five times its volume of dry toluene with vigorous agitation in the flask. The purified tannic acid is precipitated immediately in the toluene. Filter by suction through a sintered glass funnel and wash the product with dry toluene, stirring the solids with a stream of toluene to effect complete removal of ethyl acetate. Continue the suction until practically all the toluene has been removed. Dry this purified tannic acid in a vacuum oven at about 40°C, and place in a tightly stoppered bottle.

4.4.2.2 Apparatus.

Soxhlet extraction apparatus with all glass connections (condenser, Soxhlet extractor, and 250 ml Erlenmeyer flask).

Volumetric flask - 100 ml.

Test tubes - 18 by 150 mm.

Test tube basket, for 18 by 150 mm test tubes.

Water bath, maintained at 100°C.

Klett-Summerson photoelectric colorimeter or equal.

4.4.2.3 Procedure for analysis. Extract a quantity of the sample comprising an accurately measured cross-section of treated paper free of any colored markings (1 square foot (0.0929 m^2) should contain at least 40 mg of pure piperonyl butoxide). Cut the sample into 5 mm by 75 mm strips and extract for 5 hours in a 250 ml Soxhlet extraction apparatus with acetone. A wire screen is placed in the bottom of the extraction tube and the strips are placed upon this screen. After extraction, the acetone is removed from the extract by evaporating the solvent on a hot water bath, heating no longer than is necessary to remove the solvent. The residue is dissolved in deodorized kerosene, then transferred to a 100 ml volumetric flask and the solution made to volume. The size of the sample and the volume of the final dilution may be varied, but should be so adjusted as to give between 25 and 50 micrograms (ug) of piperonyl butoxide per 0.1 ml aliquot of the test solution. Exactly 0.1 ml of solution of the

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test sample and standard is placed in 18 by 150 mm test tubes. To these is added from a burette exactly 5.0 ml of the tannic acid reagent. Following addition of reagent, the tubes are shaken vigorously by hand for 1 minute. Reagent withdrawal from burette to sample tube should be timed to encompass at least a 60 second interval so that accurate meniscus recovery can be accomplished. The standard, the test sample, and a blank of 0.1 ml of deodorized kerosene, are treated simultaneously in the same manner. The test tubes are placed in a test tube basket and the basket placed in a vigorously boiling water bath for 5 to 6 minutes while the blue color develops. At the end of this period, the basket is removed and the tubes allowed to cool to room temperature. The samples are then transferred to colorimeter tubes and read in the usual manner, using a color filter having a transmission range of 625 to 635 mu in a photoelectric colorimeter.

4.4.2.4 Calculation of results. The readings for the blank of deodorized kerosene are subtracted from the readings of both the unknown and the standard. The results are then calculated by a direct proportion between the unknown and the standard. For example, if a standard prepared by extracting a quantity of the original insecticide in the same manner accorded the test samples, and which from its analysis is calculated to contain 50 mg of piperonyl butoxide per 100 ml, gives an absorbency of 0.150 and the unknown solution gives an absorbency of 0.141 (both after correction for the blank), the calculation is as follows:

$$\frac{X}{50} = \frac{0.141}{0.150}$$

X = 47 mg pure piperonyl butoxide per square foot. ($0.0929\ m^2$)

4.4.3 Method for analysis of sulfoxide on treated paper.

4.4.4.1 Reagents.

Sulfoxide concentrate (see 6.4).

Standard solutions of sulfoxide - Prepare standard solutions which contain 10, 20, 30, 40, 50, 60 and 70 mg of sulfoxide per 100 ml of acetonitrile.

Gallic acid - A.C.S. grade, purified. Recrystallized from hot water.

Acetonitrile - A.C.S. grade, redistilled.

Sulfuric acid - A.C.S. grade, concentrated.

Acetone - A.C.S. grade.

Ethyl ether - A.C.S. grade.

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4.4.4.2 Apparatus.

Micro burette.

Test tubes - 13 by 100 mm.

Spectrophotometer - Beckman DU or equal.

4.4.3.3 Preparation of standard curve. Place 10 mg of gallic acid into each of seven dry 13 by 100 mm test tubes. Add a 0.1 ml aliquot of each standard solution of sulfoxide into individual test tubes, containing the gallic acid, by means of a micro burette. The resulting test tubes contain 10, 20, 30, 40, 50, 60 and 70 micrograms of sulfoxide respectively. Cautiously add 3 ml of sulfuric acid down the side of each test tube so that it layers. After 1 minute, the acid and standard sulfoxide solution are thoroughly mixed and allowed to stand 2 hours for maximum color development. Transfer the samples to cuvettes and read the absorbency in a Beckman DU or equivalent spectrophotometer at 660 mu. A blank determination is made in the same manner for all the reagents used and the reading subtracted from the readings obtained on the standard. A standard curve is made plotting micrograms of sulfoxide against absorbence readings.

4.4.3.4 Determination of sulfoxide on treated paper. Cut into suitable size strips, 0.5 square foot (0.047 m^2) of the paper sample which is free of any colored markings, and place directly into the Soxhlet extractor and extract for 3 hours with 125 ml of acetone. Remove the extract and evaporate to dryness on a steam bath. Transfer the contents of the flask to a 250 ml separatory funnel with 25 ml of distilled water and 50 ml of ethyl ether. Shake vigorously for one minute and allow to stand 5 to 10 minutes for phase separation. Transfer the aqueous layer to a second 250 ml separatory funnel and wash with 10 ml of ethyl ether. Continue extracting and draining aqueous layer, using 10 ml portion of ethyl ether for third and fourth extraction. Combine the four ether extracts and evaporate to dryness on a steam bath. Transfer the dried and cooled residue to a 10 ml volumetric flask with acetonitrile and make to volume. From this point proceed as in the preparation of the standard curve. Paper of the same origin as the sample, but not treated with insecticide, should be run through this procedure and used as a correction blank. Refer the absorbence found to the standard curve and calculate milligrams of sulfoxide per square foot (0.093 m^2) of paper.

5. PACKAGING

This section is not applicable to this specification.

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6. NOTES

6.1 Intended use. Insect resistant treatment is intended for use on paper for package wrapping and multiwall bags for subsistence items (see Federal Specification UU-S-48 for Sacks, Shipping, Paper) to afford protection against insect penetration. The treated side of the paper shall be on the exterior surface of the outer ply of the wrap or pack.

* 6.1.1 Limitation. Dependent upon storage conditions of the end item this insect resistant treatment should be effective for 9 to 12 months. Maximum effectiveness is achieved by storage in the dark. The treated paper should not be used in direct contact with any edible subsistence item nor shall the paper come in direct contact with cooking or eating utensils. Furthermore, at least two plies of untreated paper (or of any material which can be shown to be an effective barrier to the treatment chemicals) should be placed between the treated paper and the subsistence item. Examples are as follows:

(a) Flour packaged in four-ply bags with the exterior surface of the outer ply treated.

(b) Subsistence packaged in bulk such as macaroni, using package interliners consisting of two plies of untreated paper and an outer ply of treated paper. Any application requiring closer contact between treated paper and subsistence than indicated in the foregoing should be referred for approval to the Medical Authority of the department effecting the procurement.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title number and date of this specification
(b) Title number and date of the untreated paper specification (see 3.1)

6.3 Loss in strength of treated paper. The physical requirements of insect resistant treated papers are lower than those for the untreated papers. Procurement documents of specifications requiring the use of treated paper should specify the minimum physical requirements acceptable for the treated paper. The reduction in the physical strength of paper through the addition of insect resistant coating should average less than 10 percent.

6.4 Sulfoxide concentrate. Sulfoxide concentrate specified in 4.4.4.1 can be obtained from S. B. Penick and Company, 100 Church Street, New York, NY 10007.

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6.5 Alternate provisions. Changes in this specification or modifications at time of use in procurement should be cleared with the U. S. Army Natick Research and Development Command, Natick, Massachusetts 01760.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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